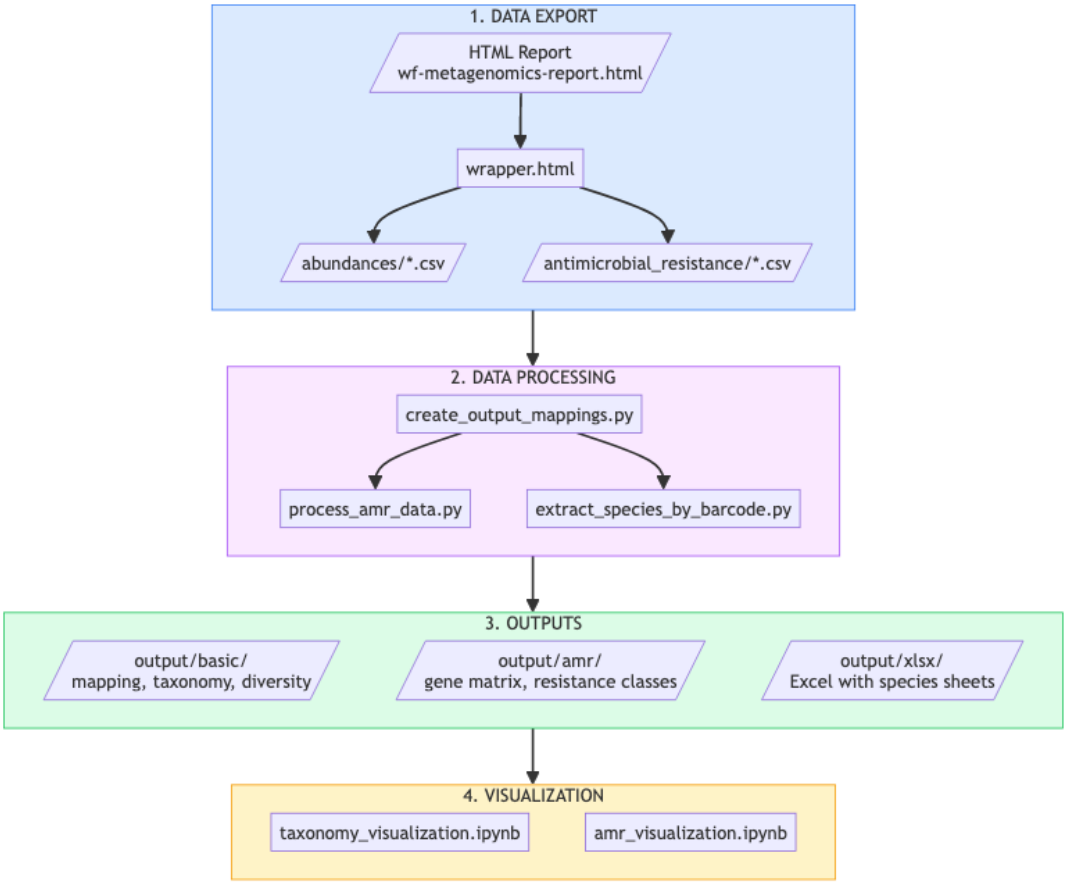
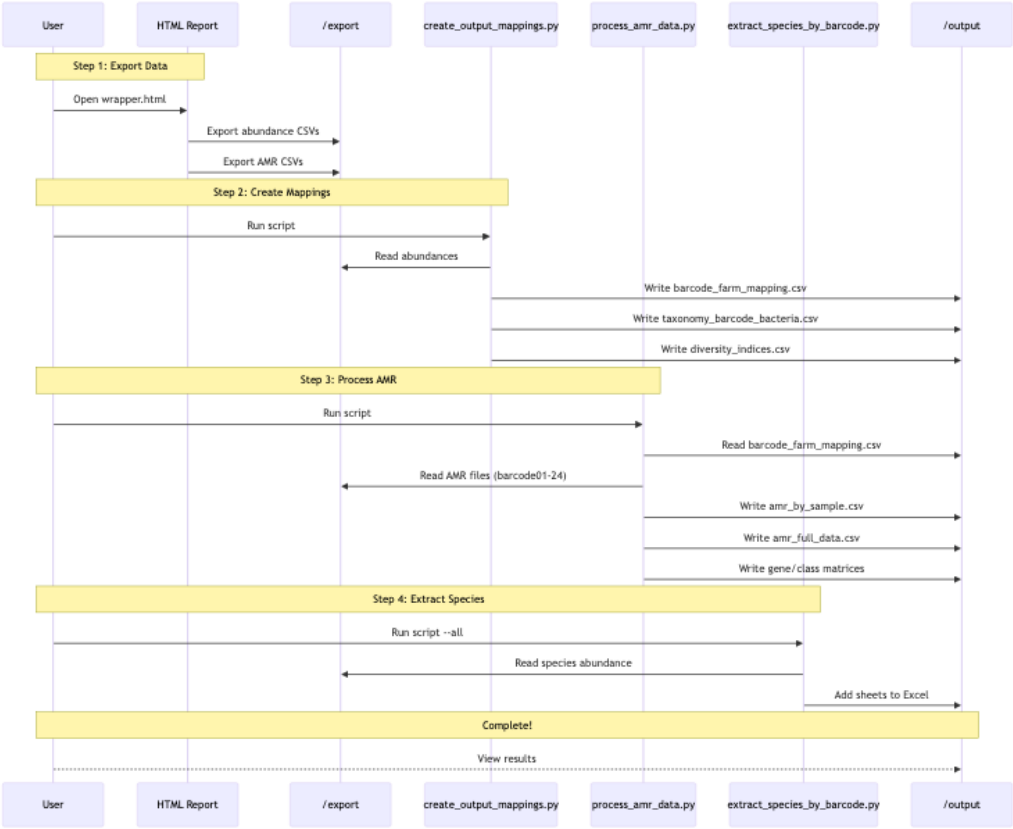
Workflow Documentation

This document provides visual documentation of the bio-phagent data pipeline.

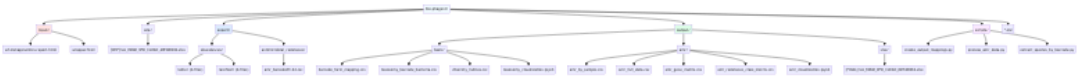
Pipeline Overview



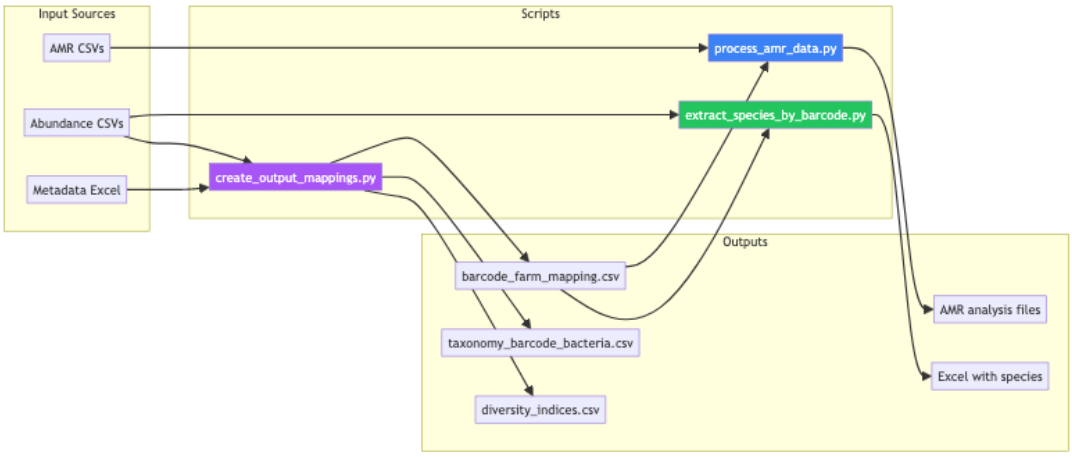
Data Flow Sequence



File Structure



Script Dependencies



Input/Output Schemas

Abundance Table Format

```
| species      | barcode01 | barcode02 | ... | barcode24 |
phylum   | class     | ... |
|--------------|-----------|-----------|-----|-----------|---------
| E. coli      | 15234     | 12456     | ... | 8923      |
Pseudo.. | Gamma..   | ... |
```

AMR File Format

```
| Gene         | ReadID                                | Coverage %
| Identity % | Resistance          |
|--------------|---------------------------------------|-----------
| aph(3'')-Ib  | b963a3b5-5373-43f2-8d97-47fbacb269fa | 98.76
| 96.56      | Streptomycin        |
| tetA         | c1d2e3f4-...                          | 95.43
| 98.21      | Tetracycline;Chlor.. |
```

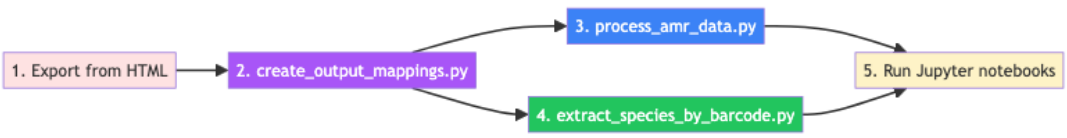
Output: barcode_farm_mapping.csv

```
| barcode   | sample_id | farm_type | farm_id     | sample_type |
oblast              |
|-----------|-----------|-----------|-------------|-------------|--
| barcode01 | 1.1.      | Poultry   | poultry #1  | Native      |
Zaporizhzhia Oblast |
| barcode02 | 1.1. n    | Poultry   | poultry #1  | Enriched    |
Zaporizhzhia Oblast |
```

Output: amr_by_sample.csv

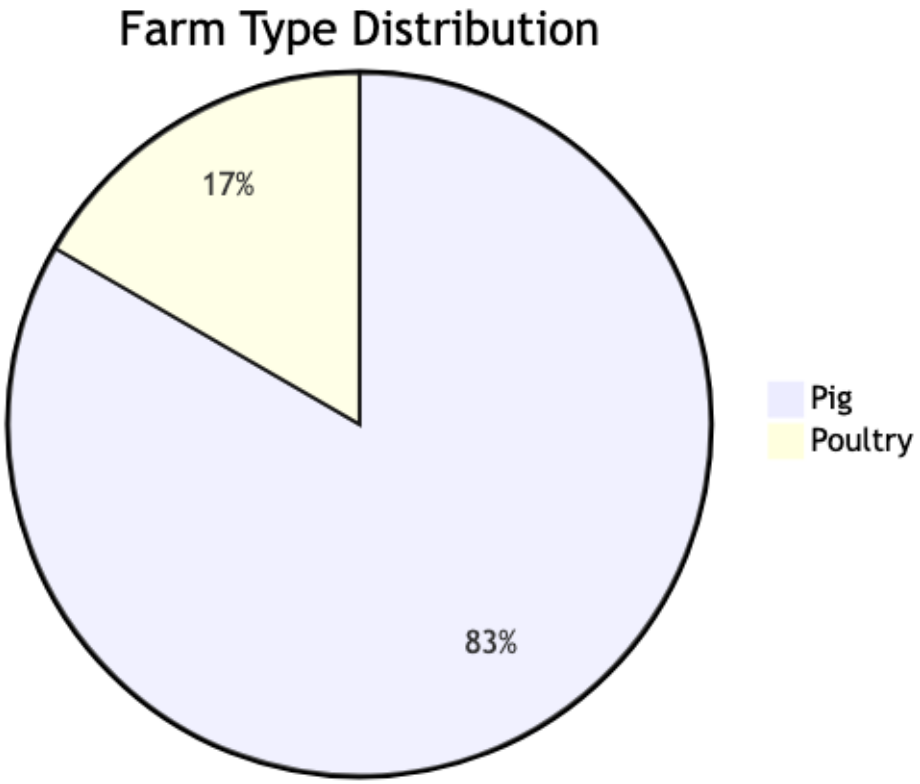
```
| barcode   | farm_type | total_gene_detections | unique_reads |
unique_genes | unique_resistance_classes |
|-----------|-----------|-----------------------|--------------|----
| barcode01 | Poultry   | 247                   | 211          |
13           | 27                        |
| barcode02 | Poultry   | 2727                  | 2586         |
46           | 22                        |
```

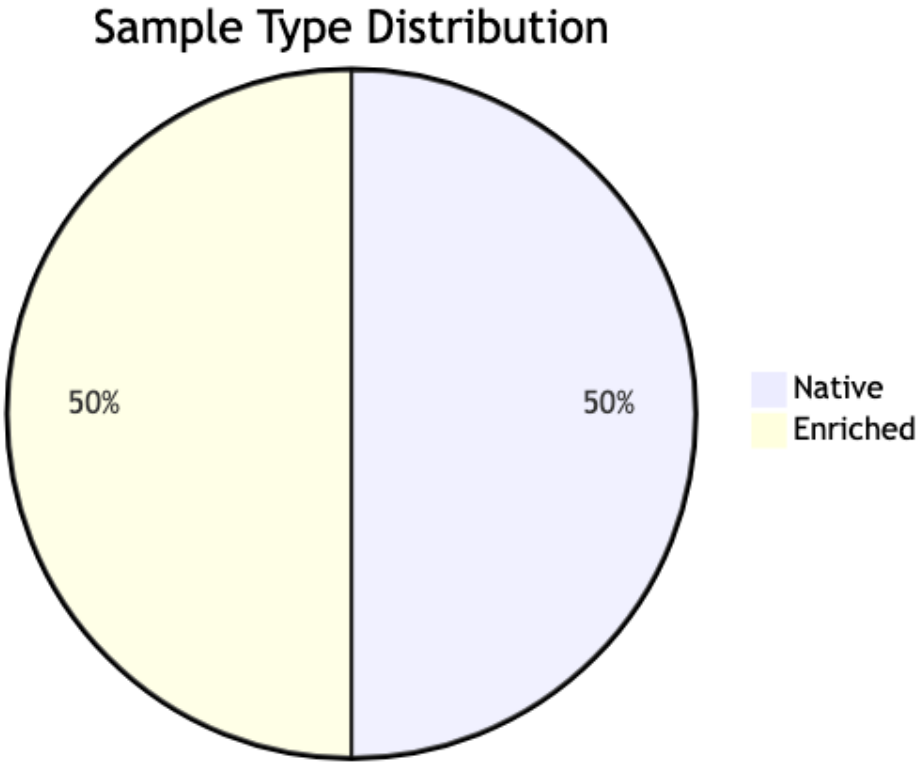
Execution Order



Note: Steps 3 and 4 can run in parallel after step 2.

Sample Distribution




Sample Type Distribution
50%
50%
Native
Enriched